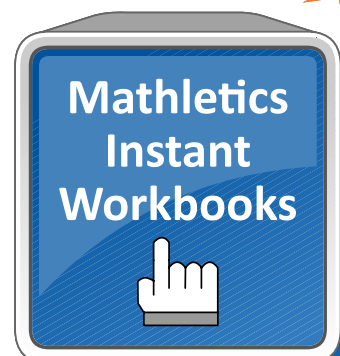
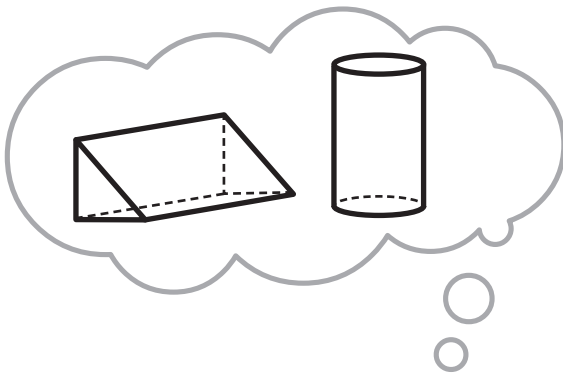


MATHLETICS

Surface Area and Volume

Student Book - Series J-1



Surface area and volume

Student Book - Series J

Contents

Topics

Date completed

Topic 1 - Surface area of a right prism

__/__/__

Topic 2 - Surface area of a right cylinder

__/__/__

Topic 3 - Volume of a right prism

__/__/__

Topic 4 - Volume of a right cylinder

__/__/__

Topic 5 - Volume of a pyramid

__/__/__

Topic 6 - Volume of a right cone

__/__/__

Topic 7 - Volume of a sphere

__/__/__

Topic 8 - Applications of area and volume

__/__/__

Practice Tests

Topic 1 - Topic test A

__/__/__

Topic 2 - Topic test B

__/__/__

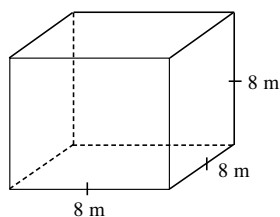
Author of The Topics and Topic Tests: AS Kalra

Surface area and volume

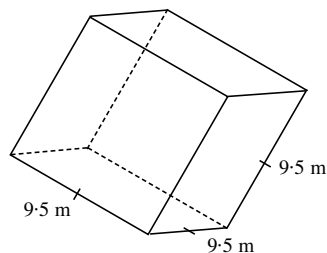
Topic 1: Surface area of a right prism

QUESTION 1 Find the surface area of each cube.

a

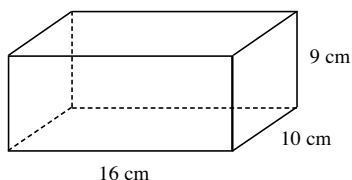


b

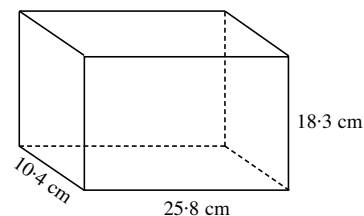


QUESTION 2 Find the surface area of each rectangular prism.

a

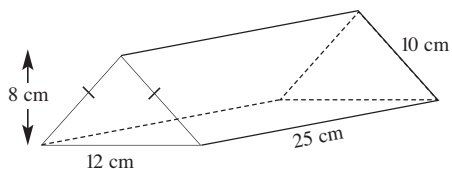


b

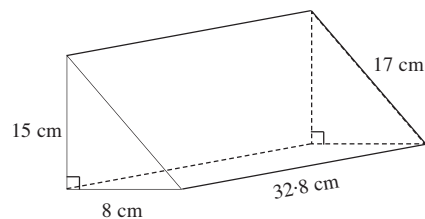


QUESTION 3 Find the surface area of each triangular prism.

a

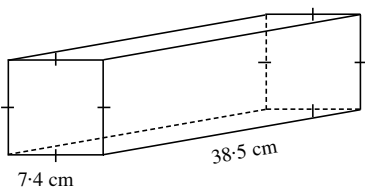


b

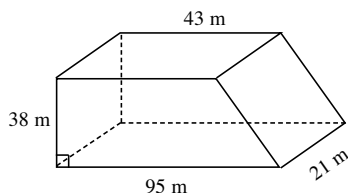


QUESTION 4 Find the surface area of each shape.

a



b



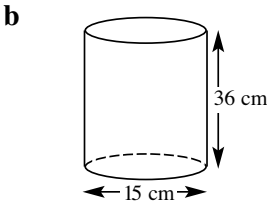
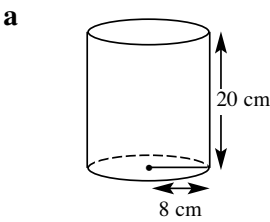
Surface area and volume

Topic 2: Surface area of a right cylinder

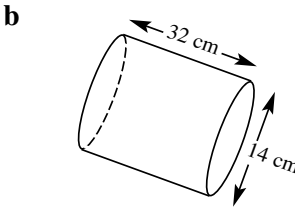
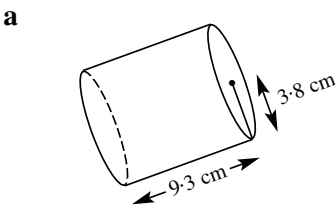
QUESTION 1 For each cylinder, find the following correct to two decimal places.

i the area of a circular base

ii the area of the curved surface



QUESTION 2 Find the curved surface area of each cylinder in terms of π .

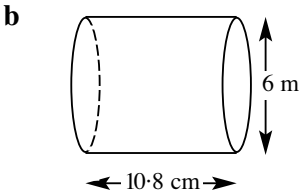
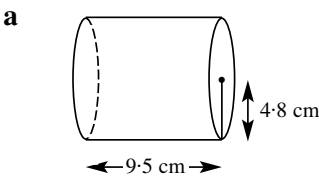


QUESTION 3 For each cylinder, find the following correct to three significant figures.

i the combined area of the two circular ends

ii the area of the curved surface

iii the total surface area



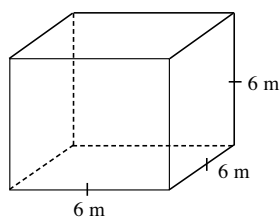
QUESTION 4 Find the total surface area of the outside of a pipe 20 m long with an outer radius 0.75 m (a pipe does not have any ends). Give your answer correct to one decimal place.

Surface area and volume

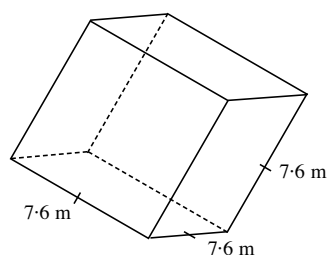
Topic 3: Volume of a right prism

QUESTION 1 Find the volume of each cube.

a

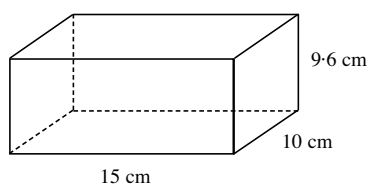


b

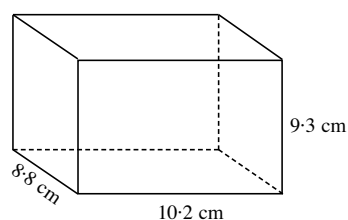


QUESTION 2 Find the volume of each rectangular prism.

a

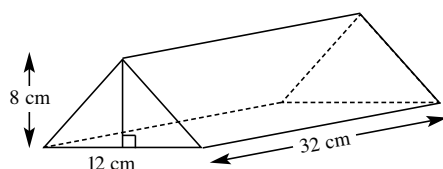


b

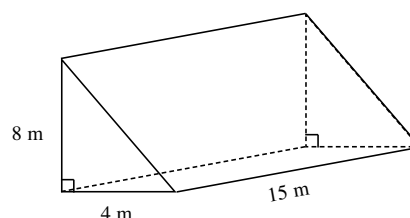


QUESTION 3 Find the volume of each triangular prism.

a

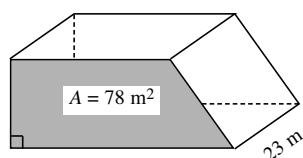


b

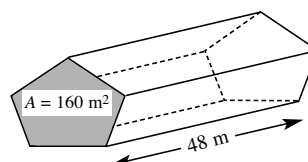


QUESTION 4 Find the volume of each prism, given the area of the shaded face.

a



b



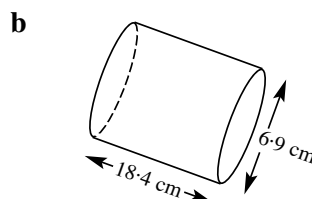
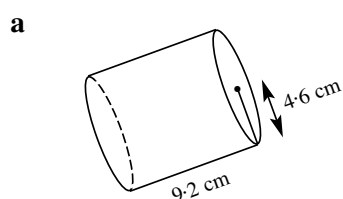
Surface area and volume

Topic 4: Volume of a right cylinder

QUESTION 1 Find the volume of each cylinder correct to two significant figures.

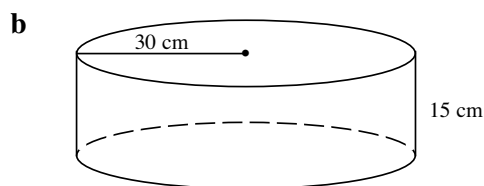
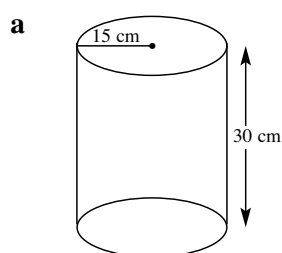
- a radius 6 cm and height 20 cm _____
- b radius 9.6 cm and height 18 cm _____
- c radius 20.8 cm and height 30.4 cm _____
- d radius 4.6 m and height 15.6 m _____

QUESTION 2 Find the volume of each correct to two decimal places.

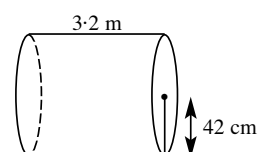


QUESTION 3 Find the volume in cubic centimetres correct to one decimal place of a soft drink can with height 115 mm and radius 30 mm.

QUESTION 4 Which of the following cylinders has the larger volume?



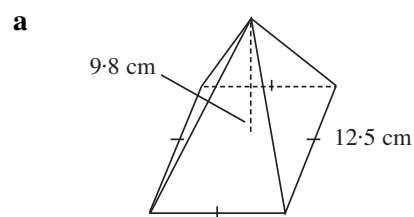
QUESTION 5 Find the volume of this cylinder in cubic metres correct to three significant figures.

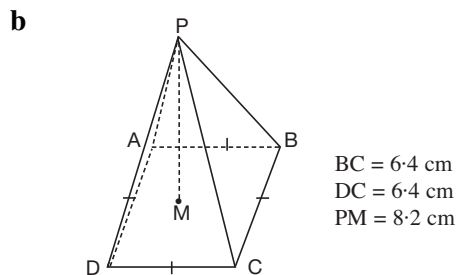


Surface area and volume

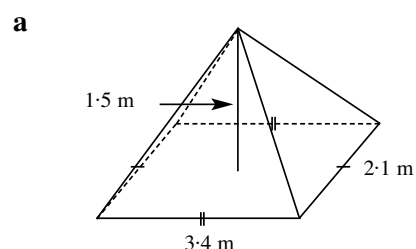
Topic 5: Volume of a pyramid

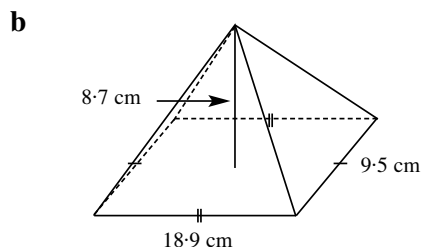
QUESTION 1 Calculate the volume of the following square pyramids correct to one decimal place.



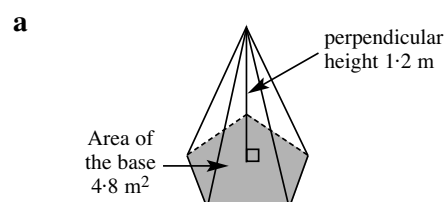


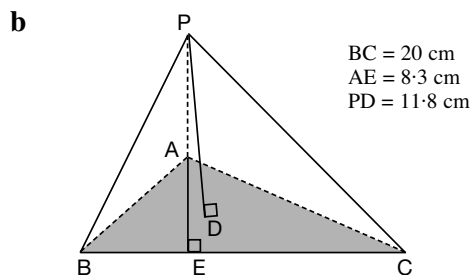
QUESTION 2 Calculate the volume of the following rectangular pyramids correct to two decimal places.





QUESTION 3 Calculate the volume of the following pyramids correct to one decimal place.



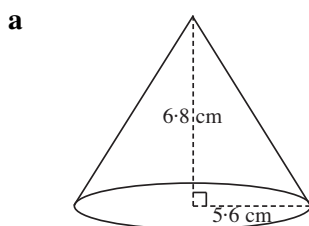


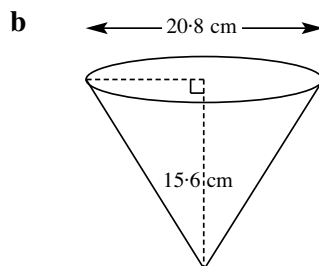
QUESTION 4 The area of the base of an octagonal pyramid is 225 cm² and its height is 16.4 cm. Find its volume.

Surface area and volume

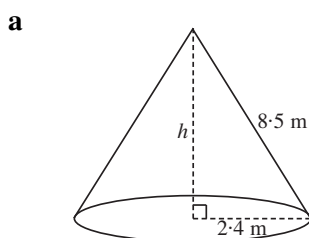
Topic 6: Volume of a right cone

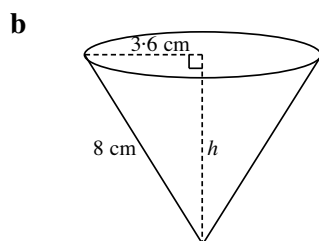
QUESTION 1 Find the volume of the following cones correct to one decimal place.





QUESTION 2 Find the volume of the following cones correct to two decimal places.





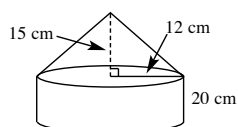
QUESTION 3

a A cone has a base radius of 12 cm and a height of 20 cm. Find its volume.

b Find the volume of a cone of height 8.4 cm and base diameter 6.2 cm.

c Find the volume of a cone that has a slant height of 17 cm and base diameter of 16 cm.

QUESTION 4 Find the volume of the solid.



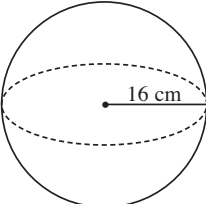
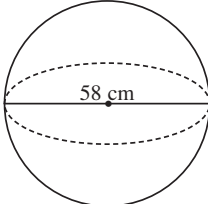
Surface area and volume

Topic 7: Volume of a sphere

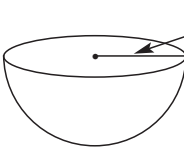
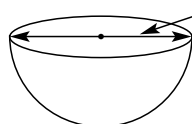
QUESTION 1 Find the volume, correct to one decimal place, of a sphere with the following.

- | | | |
|------------------------|-------------------------|-------------------------|
| a radius 7 cm | b diameter 18 cm | c radius 25 mm |
| _____ | _____ | _____ |
| _____ | _____ | _____ |
| _____ | _____ | _____ |
| d diameter 28 m | e diameter 63 cm | f radius 12.4 km |
| _____ | _____ | _____ |
| _____ | _____ | _____ |
| _____ | _____ | _____ |

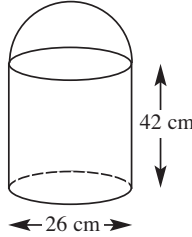
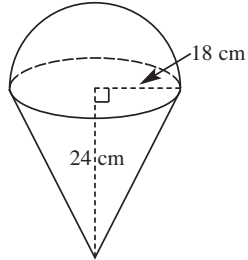
QUESTION 2 Calculate the volume of the following spheres correct to one decimal place.

- | | | |
|----------|---|-------|
| a |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| b |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |

QUESTION 3 Calculate the volume of the following hemispheres correct to one decimal place.

- | | | |
|----------|--|-------|
| a |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| b |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |

QUESTION 4 Calculate the volume of the following solids correct to one decimal place.

- | | | |
|----------|--|-------|
| a |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| b |  | _____ |
| | | _____ |
| | | _____ |
| | | _____ |
| | | _____ |

Surface area and volume

Topic 8: Applications of area and volume

QUESTION 1 Complete the following.

a $1 \text{ cm}^3 =$ _____ mL b $1000 \text{ cm}^3 =$ _____ L c $1 \text{ m}^3 =$ _____ L

QUESTION 2 A pot has a volume of $15\,000 \text{ cm}^3$. How many litres of water can it hold?

QUESTION 3 The radius of the earth is approximately 6400 km.

- a Given that the surface area of a sphere is $4\pi r^2$, find the surface area in square kilometres. b Calculate the volume correct to four significant figures.

QUESTION 4 A rectangular roof is 28 m long and 12 m wide.

- a What volume of water will fall on the roof if we receive 20 mm of rain?

- b A tank catches all the rain that falls on the roof. How many litres of water will flow into the tank from 20 mm of rain?

- c The tank holds 35 000 litres. How much rain would need to fall to fill the tank if it is empty and only catches rain from the above roof?

QUESTION 5 A rectangular swimming pool with uniform depth is 30 metres long, 8 metres wide and 2.8 metres deep. It is to be tiled. Calculate the following.

- a the cost of tiling it at \$53 per square metre b its capacity in litres

Surface area and volume

Topic Test

PART A

Time allowed: 15 minutes

Total marks = 15

		Marks
1	The diameter and radius of a circle are related as (A) $rd = 2$ (B) $r = 2d$ (C) $d = 2r$ (D) $\frac{r}{d} = 2$	1
2	The circumference of a circle is given by the formula (A) $C = \frac{2\pi}{r}$ (B) $C = 2\pi r$ (C) $C = 2\pi d$ (D) $C = \frac{2\pi}{d}$	1
3	The area of a circle is given by the formula (A) $A = \frac{\pi}{r^2}$ (B) $A = \frac{\pi}{d^2}$ (C) $A = \pi r^2$ (D) $A = \pi d^2$	1
4	The volume of a cylinder with radius r and height h equals (A) $V = \pi^2 rh$ (B) $V = \pi rh^2$ (C) $V = \frac{1}{3}\pi r^2 h$ (D) $V = 2\pi rh$	1
5	A semi-circle equals (A) a full circle (B) half a circle (C) a quarter of a circle (D) a third of a circle	1
6	A quadrant is (A) $\frac{3}{4}$ of a circle (B) $\frac{1}{2}$ of a circle (C) $\frac{1}{3}$ of a circle (D) $\frac{1}{4}$ of a circle	1
7	The shaded area in the figure is called a  (A) semi-circle (B) segment (C) chord (D) sector	1
8	How many square centimetres are there in one square metre? (A) 100 (B) 1000 (C) 10 000 (D) 100 000	1
9	A rectangular prism is 10 cm long, 8 cm wide and 4 cm high. Its surface area is (A) 152 cm ² (B) 304 cm ² (C) 320 cm ² (D) 640 cm ²	1
10	Give the total surface area in cm ² correct to one decimal place of a closed cylinder with dimensions of radius 6 cm and height 15 cm. (A) 226.2 cm ² (B) 565.5 cm ² (C) 791.7 cm ² (D) 678.6 cm ²	1
11	A cube has a volume of 729 cm ³ . Find the length of each side of the cube. (A) 6 cm (B) 9 cm (C) 18 cm (D) 27 cm	1
12	A cylinder has height 9 m and radius 6 m. Its volume is closest to (A) 113 m ³ (B) 452 m ³ (C) 2036 m ³ (D) 1018 m ³	1
13	The volume of a rectangular pyramid with base area of 75 cm ² and vertical height of 8 cm is (A) 200 cm ³ (B) 400 cm ³ (C) 600 cm ³ (D) 800 cm ³	1
14	The volume of a cone with diameter 12 cm and height 8.5 cm is closest to (A) 320 cm ³ (B) 961 cm ³ (C) 1282 cm ³ (D) 3845 cm ³	1
15	The volume of a sphere of diameter 24 cm is closest to (A) 1810 cm ³ (B) 7238 cm ³ (C) 14 476 cm ³ (D) 57 906 cm ³	1

Total marks achieved for PART A

15

Surface area and volume

Topic Test

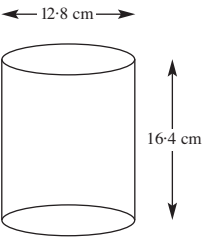
PART B

Time allowed: 15 minutes

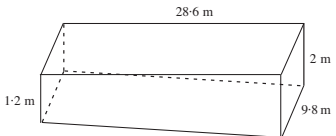
Total marks = 15

Question 1 For this closed cylinder, find the following correct to two decimal places.

- a the area of a circular base
- b the area of both the circular bases
- c the area of its curved surface
- d the total surface area
- e the volume of this closed cylinder



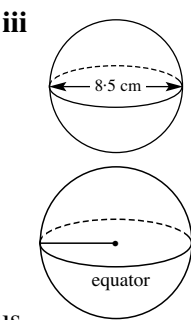
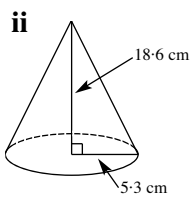
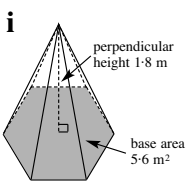
Question 2 A swimming pool has the shape of a trapezoidal prism as shown.



- a Find the volume of the pool in m^3 .
- b What is the capacity of the pool in kilolitres?
- c The mass of 1 kL of water is 1 t. How many tonnes of water are in the pool?
- d Tom treats this pool with a chlorine product to prevent the growth of algae. The recommended dose is 4 g of chlorine for each 100 L of water. How much chlorine must Tom place in the pool?
- e Over a period of hot weather, the level of the pool dropped by 50 cm. The pool was originally full. What volume of water, in litres, evaporated?

Question 3

- a Find the volume of each of the following correct to two decimal places.



- b The circumference of the earth at the equator is about 40 000 km.
 - i Use the formula $C = 2\pi r$ to find the radius of the earth correct to the nearest 100 km.
 - ii Use this radius to find the volume of the earth correct to two significant figures. Write your answer in scientific notation.

Marks

1
1
1
1
1

1
1
1

1
1

1
1

1
1

1

Total marks achieved for PART B

15